Mathematics

Name:

TEACHER

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S5EN….

FIRST
SEMESTER

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1. Solve the following using three Methods (factors, formula, complete square)

$x^2 + 10x - 11=0$	$x^2 + 10x - 11=0$	$x^2 + 6x - 7=0$

2. Solve the following using three Methods (factors, formula, complete square)

$-x^2 - 10x + 11 = 0$	$-x^2 - 10x + 11 = 0$	$-x^2 - 10x + 11 = 0$

3. Given the function $f(x) = x^2 + 10x - 11$

i. Determine the point where f(x) crosses the y-axis.

ii. Determine the points where f(x) crosses the x-axis

iii. Complete the square to determine the coordinates of the minimum point of the curve

iv. Sketch the graph

4. Given the function $f(x) = -x^2 - 10x + 11$

ii. Determine the point where f(x) crosses the y-axis.

iii. Determine the points where f(x) crosses the x-axis

iv. Complete the square to determine the coordinates of the maximum point of the curve

v. Sketch the graph

5. Solve the system of simultaneous equations for x and y

$$x + 2y = -3$$
$$x^2 + y^2 = 5$$

6. Sasha is playing golf. The following function models the height of the ball,h after the number of seconds t

$$h(t) = -16x^2 + 48t$$

i) Sketch the function.
ii) After how many seconds will the ball reach it's highest point.
iii) What is it's highest altitude
iv) After how many seconds will the ball hit the ground again.

7. Solve the following disguised quadratics

a) $x - 3 = \frac{10}{x}$

$b)$ $x^4 - 26x^2 + 25 = 0$

c) $5^{2x} - 26(5^x) + 25 = 0$